

1. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: [0, \frac{3\pi}{4}]$ defined by

$$\vec{r}(t) = (3 \sin(2t), 2 \cos(2t)) \quad \text{with} \quad 0 \leq t \leq \frac{3\pi}{4}.$$

- (i) The curve $\mathcal{C}$ is the arc of ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ lying in the first, third and fourth quadrants.
- (ii) The point $(\frac{3\sqrt{2}}{2}, -\sqrt{2})$ corresponds to $t = \frac{3\pi}{8}$ and hence the direction of the tangent line to $\mathcal{C}$ at the point $(\frac{3\sqrt{2}}{2}, -\sqrt{2})$ is given by $\vec{r}'(3\pi/8)$.

$$\vec{r}'(t) = (6 \cos(2t), -4 \sin(2t)) \quad \Rightarrow \quad \vec{r}'(3\pi/8) = (-3\sqrt{2}, -2\sqrt{2}).$$

Thus, the parametric equations of the tangent line to $\mathcal{C}$ at the point $(\frac{3\sqrt{2}}{2}, -\sqrt{2})$ are

$$x = \frac{3\sqrt{2}}{2} - 3\lambda\sqrt{2}, \quad y = -\sqrt{2} - 2\lambda\sqrt{2}, \quad \lambda \in \mathbb{R}.$$

2. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}(t) = (2 \sin(t\sqrt{2}), t, 2 \cos(t\sqrt{2}))$ with $0 \leq t \leq \frac{\pi}{\sqrt{2}}$.

The points $(2, \frac{\pi}{2\sqrt{2}}, 0)$ and $(0, \frac{\pi}{\sqrt{2}}, -2)$ correspond to $t = \frac{\pi}{2\sqrt{2}}$ and $t = \frac{\pi}{\sqrt{2}}$ respectively. So,

$$L = \int_{\frac{\pi}{2\sqrt{2}}}^{\frac{\pi}{\sqrt{2}}} |\vec{r}'(t)| dt.$$

$$\vec{r}'(t) := (2\sqrt{2} \cos(t\sqrt{2}), 1, -2\sqrt{2} \sin(t\sqrt{2})) \quad \Rightarrow \quad |\vec{r}'(t)| = 3.$$

Thus,

$$L = \frac{3\pi}{2\sqrt{2}}.$$

3. Given two real numbers a and b both different from 0.

- (i) Find the arc length parameterization $\vec{\gamma}(s)$ of the curve described by $\vec{r}(t) = (a \cos t, a \sin t, bt)$ with $0 \leq t \leq 2\pi$.

Find arc length function $s = \int_0^t |\vec{r}'(u)| du = t\sqrt{a^2 + b^2}$. So the arc length parametrization is

$$\boxed{\vec{\gamma}(s) = \vec{r}(s/\sqrt{a^2 + b^2}) = \left(a \cos \left(\frac{s}{\sqrt{a^2 + b^2}} \right), a \sin \left(\frac{s}{\sqrt{a^2 + b^2}} \right), \frac{bs}{\sqrt{a^2 + b^2}} \right)}$$

with $0 \leq s \leq 2\pi\sqrt{a^2 + b^2}$.

- (ii) Show that $\vec{\gamma}'(s)$ is a unit vector.

We find that

$$\vec{\gamma}'(s) = \left(-\frac{a}{\sqrt{a^2 + b^2}} \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right), \frac{a}{\sqrt{a^2 + b^2}} \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right), \frac{b}{\sqrt{a^2 + b^2}} \right).$$

So,

$$\begin{aligned} |\vec{\gamma}'(s)|^2 &= \frac{a^2}{a^2 + b^2} \sin^2\left(\frac{s}{\sqrt{a^2 + b^2}}\right) + \frac{a^2}{a^2 + b^2} \cos^2\left(\frac{s}{\sqrt{a^2 + b^2}}\right) + \frac{b^2}{a^2 + b^2} \\ &= \frac{a^2}{a^2 + b^2} \left[\sin^2\left(\frac{s}{\sqrt{a^2 + b^2}}\right) + \cos^2\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \right] + \frac{b^2}{a^2 + b^2} \\ &= \frac{a^2}{a^2 + b^2} + \frac{b^2}{a^2 + b^2} = \frac{a^2 + b^2}{a^2 + b^2} = 1. \end{aligned}$$

- (iii) Find the scalar product $\vec{\gamma}'(s) \cdot \vec{\gamma}''(s)$.

We find that

$$\vec{\gamma}''(s) = \left(-\frac{a}{a^2 + b^2} \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right), -\frac{a}{a^2 + b^2} \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right), 0 \right).$$

So,

$$\begin{aligned} \vec{\gamma}'(s) \cdot \vec{\gamma}''(s) &= \frac{a^2}{(a^2 + b^2)^{3/2}} \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \\ &\quad - \frac{a^2}{(a^2 + b^2)^{3/2}} \sin\left(\frac{s}{\sqrt{a^2 + b^2}}\right) \cos\left(\frac{s}{\sqrt{a^2 + b^2}}\right) = 0. \end{aligned}$$

- (iv) Where you expecting this value? YES! Since, $\vec{\gamma}'(s)$ is a unit vector i.e., $|\vec{\gamma}'(s)| = 1$. In fact, if we differentiate both sides of the identity

$$\vec{\gamma}'(s) \cdot \vec{\gamma}'(s) = 1,$$

we obtain that

$$\begin{aligned} \vec{\gamma}''(s) \cdot \vec{\gamma}'(s) + \vec{\gamma}'(s) \cdot \vec{\gamma}''(s) &= 0 \Leftrightarrow 2\vec{\gamma}'(s) \cdot \vec{\gamma}''(s) = 0 \\ &\Leftrightarrow \vec{\gamma}'(s) \cdot \vec{\gamma}''(s) = 0 \Leftrightarrow \vec{\gamma}'(s) \perp \vec{\gamma}''(s). \end{aligned}$$

4. Consider the curve $\mathcal{C}$ described by the vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) = (e^t \sin t, e^{2t} \cos t, 2), \quad t \in [0, 2\pi].$$

- (a) $\vec{r}'(t) = (e^t(\sin t + \cos t), e^{2t}(2 \cos t - \sin t), 0)$. So, $\vec{r}'(\pi/2) = (e^{\pi/2}, -e^\pi, 0)$.
 (b) The point $(e^{\pi/2}, 0, 2)$ corresponds to $t = \frac{\pi}{2}$. So, the direction of the tangent line to the curve $\mathcal{C}$ at the point $(e^{\pi/2}, 0, 2)$ is given by $\vec{r}'(\frac{\pi}{2})$. Thus, the tangent line to $\mathcal{C}$ at the point $(e^{\pi/2}, 0, 2)$ has parametric equations

$$x = e^{\pi/2} + \lambda e^{\pi/2}, \quad y = -\lambda e^\pi, \quad z = 2, \quad \lambda \in \mathbb{R}.$$

- (c) The tangent line to the curve $\mathcal{C}$ at the point corresponding to $\vec{r}(t)$ is parallel to the y -axis if its direction $\mathbf{r}'(t)$ is parallel to the vector $\mathbf{j} = (0, 1, 0)$, that is $e^t(\sin t + \cos t) = 0$ which implies $\sin t = -\cos t$. The two values of t are $\frac{3\pi}{4}$ and $\frac{7\pi}{4}$. These correspond to the points $\left(\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}, -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}, 2\right)$ and $\left(-\frac{\sqrt{2}}{2}e^{\frac{7\pi}{4}}, -\frac{\sqrt{2}}{2}e^{\frac{7\pi}{4}}, 2\right)$ respectively.

5. Given the curve $\mathcal{C}$ described by the vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) = (2 \sin(2t), t, 2 \cos(2t)) \quad t \in [0, 2\pi].$$

- (i) $\vec{r}'(t) = (4 \cos(2t), 1, -4 \sin(2t))$ $|\vec{r}'(t)| = \sqrt{16 \cos^2(2t) + 1 + 16 \sin^2(2t)} = \sqrt{17}$.
The point $(\sqrt{3}, \pi/6, 1)$ corresponds to $t = \pi/6$ and $\vec{r}'(\pi/6) = (2, 1, -2\sqrt{3})$. Thus, the unit normal vector to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$ is

$$\vec{T}(\pi/6) = \frac{\vec{r}'(\pi/6)}{|\vec{r}'(\pi/6)|} = \frac{1}{\sqrt{17}}(2, 1, -2\sqrt{3}).$$

Since $|\vec{r}'(t)|$ is a constant function of t , we can use the formula of the curvature given by

$$\kappa(t) = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|}.$$

$\vec{T}(t) = \frac{1}{\sqrt{17}}(4 \cos(2t), 1, -4 \sin(2t))$, and hence, $\vec{T}'(t) = \frac{1}{\sqrt{17}}(-8 \sin(2t), 0, -8 \cos(2t))$.
Thus,

$$\vec{T}'(\pi/6) = \frac{1}{\sqrt{17}}(-4\sqrt{3}, 0, -4) \text{ and } |\vec{T}'(\pi/6)| = \frac{8}{\sqrt{17}}.$$

So, the curvature to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$ is

$$\kappa(\pi/6) = \frac{|\vec{T}'(\pi/6)|}{|\vec{r}'(\pi/6)|} = \frac{8}{17}.$$

The principal unit vector to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$ is

$$\vec{N}(\pi/6) = \frac{1}{\kappa(\pi/6)} \frac{\vec{T}'(\pi/6)}{|\vec{r}'(\pi/6)|} = \frac{1}{8}(-4\sqrt{3}, 0, -4).$$

- (ii) The binormal vector to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$ is

$$\vec{B}(\pi/6) = \vec{T}(\pi/6) \times \vec{N}(\pi/6) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{2}{\sqrt{17}} & \frac{1}{\sqrt{17}} & -\frac{2\sqrt{3}}{\sqrt{17}} \\ -\frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{vmatrix} = \frac{1}{8\sqrt{17}}(-4, 32, 4\sqrt{3}).$$

- (iii) The normal plane to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$ has normal vector $\vec{T}(\pi/6)$ or $\vec{r}'(\pi/6)$. Hence its cartesian equation is

$$2(x - \sqrt{3}) + (y - \pi/6) - 2\sqrt{3}(z - 1) = 0 \quad \Leftrightarrow \quad 2x + y - 2z\sqrt{3} = \pi/6.$$

The osculating plane to the curve $\mathcal{C}$ at the point $(\sqrt{3}, \pi/6, 1)$ has normal vector $\vec{B}(\pi/6)$. Hence its cartesian equation is

$$\begin{aligned} & -\frac{4}{8\sqrt{17}}(x - \sqrt{3}) + \frac{32}{8\sqrt{17}}(y - \pi/6) + \frac{4\sqrt{3}}{8\sqrt{17}}(z - 1) = 0 \\ \Leftrightarrow & -(x - \sqrt{3}) + 8(y - \pi/6) + \sqrt{3}(z - 1) = 0. \end{aligned}$$

6. Solve as in Question 3.